

Solution Requirements

1. Introduction

The success of any software application depends on clearly defining its functional and non-functional requirements. The OptiCrop system is designed to provide intelligent crop recommendations by analyzing soil nutrient values and environmental conditions using machine learning. This document specifies the requirements necessary for the successful development and deployment of the system.

2. Functional Requirements

The OptiCrop application shall provide the following functionalities:

- Accept soil nutrient values from the user.
- Accept environmental parameters such as temperature, humidity, pH, and rainfall.
- Validate the entered input values.
- Process the input data before prediction.
- Predict the most suitable crop using the trained machine learning model.
- Display the recommended crop to the user.
- Allow users to submit multiple predictions without restarting the application.

These requirements correspond to the implemented Flask application, which accepts seven inputs, processes them using the trained scaler and Logistic Regression model, and displays the predicted crop.

3. Non-Functional Requirements

The system should satisfy the following quality attributes:

Requirement	Description
Performance	Generate crop recommendations within a few seconds.
Accuracy	Provide reliable predictions using the trained machine learning model.

Reliability	Produce consistent results for valid input values.
Usability	Provide an intuitive and user-friendly interface.
Maintainability	Support future enhancements and code updates.
Scalability	Allow integration of additional features in future versions.
Availability	Ensure the application is accessible whenever required.

4. Hardware Requirements

The minimum hardware requirements are:

- Processor: Intel Core i3 or above
- RAM: 4 GB (Minimum), 8 GB Recommended
- Storage: 500 MB available space
- Display: 1366 × 768 resolution or higher
- Internet Connection: Required for deployment or online access

5. Software Requirements

Component	Technology
Programming Language	Python 3.x
Web Framework	Flask
Machine Learning Library	Scikit-learn

Data Processing	Pandas, NumPy
Visualization	Matplotlib, Seaborn
Frontend	HTML, CSS, Bootstrap
Development Environment	Jupyter Notebook
IDE	Visual Studio Code
Version Control	Git & GitHub

These technologies are reflected in the project's codebase and dependency list.

6. Input Requirements

The application requires the following user inputs:

Input Parameter	Description
Nitrogen (N)	Nitrogen content in soil
Phosphorus (P)	Phosphorus content in soil
Potassium (K)	Potassium content in soil
Temperature	Atmospheric temperature (°C)
Humidity	Relative humidity (%)

pH	Soil pH level
Rainfall	Annual rainfall (mm)

7. Output Requirements

The system generates:

- Recommended crop name.
- Prediction result displayed through the web interface.
- Fast and accurate recommendation based on user input.

8. System Constraints

The current implementation has the following limitations:

- Predictions depend on the quality of the training dataset.
- Only the supported input parameters are considered.
- Internet access is required when the application is deployed online.
- The current version focuses only on crop recommendation.

9. Assumptions

The following assumptions were considered during development:

- Users provide valid soil and environmental values.
- The dataset used for training represents realistic agricultural conditions.
- The trained machine learning model is available before prediction.
- Required software libraries are properly installed.

10. Future Requirements

Future versions of OptiCrop may include:

- Fertilizer recommendation.
- Weather API integration.
- User authentication.
- Prediction history.
- Mobile application support.
- Cloud deployment.

- Multi-language interface.

11. Conclusion

The defined solution requirements establish the foundation for the OptiCrop system. By clearly identifying functional, non-functional, hardware, and software requirements, the project ensures that the application is reliable, scalable, and capable of providing intelligent crop recommendations that support data-driven agricultural decision-making.